



Lakshya JEE 2.0 (2024)

Indefinite Integration

DPP-05

1. $\int x \cdot \log x \, dx$ is equal to:

- (1) $\frac{x^2}{4}(2\log x - 1) + c$
- (2) $\frac{x^2}{2}(2\log x - 1) + c$
- (3) $\frac{x^2}{4}(2\log x + 1) + c$
- (4) None of these

2. $\int 2x^3 \cdot e^{x^2} \, dx$

- (1) $x^2 \cdot e^{x^2} + e^{x^2} + c$
- (2) $-x^2 \cdot e^{x^2} - e^{x^2} + c$
- (3) $-x^2 \cdot e^{x^2} + e^{x^2} + c$
- (4) $x^2 \cdot e^{x^2} - e^{x^2} + c$

3. $\int 4x \cdot \cos(2-3x) \, dx$

- (1) $\frac{4x}{3} \cdot \sin(2-3x) + \frac{4}{9} \cos(2-3x) + c$
- (2) $-\frac{4x}{3} \cdot \sin(2-3x) - \frac{4}{9} \cos(2-3x) + c$
- (3) $\frac{4x}{3} \cdot \sin(2-3x) - \frac{4}{9} \cos(2-3x) + c$
- (4) $-\frac{4x}{3} \cdot \sin(2-3x) + \frac{4}{9} \cos(2-3x) + c$

4. $\int \log x^2 \, dx$ equals

- (1) $x(\log x)^2 - 2x \log x - 2x + C$
- (2) $x(\log x)^2 - 2x \log x - x + C$
- (3) $x(\log x)^2 - 2x \log x + 2x + C$
- (4) $x(\log x)^2 - 2x \log x + x + C$

5. $\int \frac{\log x}{x^2} \, dx$

- (1) $\frac{1}{2} (1 + \log x) + C$
- (2) $x(1 + \log x) + C$
- (3) $-x(1 + \log x) + C$
- (4) $-\frac{1}{x} (1 + \log x) + C$

6. $\int \frac{\sin^{-1} x \, dx}{(1-x^2)^{3/2}}$

- (1) $-\frac{x \cdot \sin^{-1} x}{\sqrt{1-x^2}} + \log \left[\sqrt{1-x^2} \right] + c$
- (2) $\frac{x \cdot \sin^{-1} x}{\sqrt{1-x^2}} + \log \left[\sqrt{1-x^2} \right] + c$
- (3) $\frac{x \cdot \sin^{-1} x}{\sqrt{1-x^2}} - \log \left[\sqrt{1-x^2} \right] + c$
- (4) $-\frac{x \cdot \sin^{-1} x}{\sqrt{1-x^2}} - \log \left[\sqrt{1-x^2} \right] + c$

7. $\int x 2^x \, dx$

8. The value of is $\int e^x (x^5 + 5x^4 + 1) \, dx$

- (1) $e^x \cdot x^5 + C$
- (2) $e^x \cdot x^5 + e^x + C$
- (3) $e^{x+1} \cdot x^5 + C$
- (4) $5x^4 \cdot e^x + C$

9. $\int e^x (\tan x + \log \sec x) \, dx$

10. Evaluate : $\int x e^{x^2} (\sin x^2 + \cos x^2) \, dx$

11. $\int \sin(\ln x) + \cos(\ln x) \, dx$

12. $\int \frac{e^x}{x} 1 + x \cdot \ln x \, dx$

13. $\int \frac{x^2 e^x}{x+2} dx$

14. The value of $\int e^x \left(\frac{1-x}{1+x^2} \right)^2 dx$ is

(1) $e^x \left(\frac{1-x}{1+x^2} \right) + C$ (2) $e^x \left(\frac{1+x}{1+x^2} \right) + C$

(3) $\frac{e^x}{1+x^2} + C$ (4) $e^x(1-x) + C$

15. If $\int e^x \left(\frac{1-\sin x}{1-\cos x} \right) dx = f(x) + \text{constant}$, then $f(x)$ is equal to

(1) $e^x \cot \left(\frac{x}{2} \right) + C$ (2) $e^{-x} \cot \left(\frac{x}{2} \right) + C$

(3) $-e^x \cot \left(\frac{x}{2} \right) + C$ (4) $-e^{-x} \cot \left(\frac{x}{2} \right) + C$

16. $\int \left(1+x - \frac{1}{x} \right) e^{x+\frac{1}{x}} dx$ is equal to

(1) $(x-1)e^{x+\frac{1}{x}} + C$ (2) $xe^{x+\frac{1}{x}} + C$

(3) $(x+1)e^{x+\frac{1}{x}} + C$ (4) $-xe^{x+\frac{1}{x}} + C$

17. If $\int e^x \left(\frac{x+2}{x+4} \right)^2 dx = \frac{f(x)e^x}{x+4} + C$ the the sum $f(x) + f^2(x) \dots \infty$ at, $x = \frac{1}{2}$ is

(1) $\frac{1}{2}$ (2) $\frac{1}{4}$

(3) 1 (4) $\frac{3}{2}$

18. If $\int x^2 e^{-2x} dx = e^{-2x} (ax^2 + bx + c) + d$, then the value of $|a/bc|$ is

(1) 1 (2) 2

(3) 3 (4) 4

19. $\int \frac{\sqrt{1-\sin x}}{1+\cos x} e^{-x/2} dx, x \in \left(0, \frac{\pi}{2} \right)$

(1) $e^{-x/2} \sin \frac{x}{2} + C$

(2) $-e^{-x/2} \sec \frac{x}{2} + C$

(3) $-e^{-x/2} \sin \frac{x}{2} + C$

(4) $-e^{-x/2} \cos \frac{x}{2} + C$

20. $\int e^{3x} \sin 2x dx$ equals

(1) $\frac{e^{3x}}{13} (3 \cos 2x + 2 \sin 2x) + C$

(2) $\frac{e^{3x}}{13} (3 \sin 2x - 2 \cos 2x) + C$

(3) $\frac{e^{3x}}{13} (-3 \sin 2x - 2 \cos 2x) + C$

(4) $\frac{e^{3x}}{13} (3 \sin 2x + 2 \cos 2x) + C$

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (1)

2. (4)

3. (4)

4. (3)

5. (4)

6. (2)

7. $\frac{2^x}{\ln 2} \left(x - \frac{1}{\ln 2} \right) + C$

8. (2)

9. $e^x \cdot \log(\sec x) + C$

10. $\frac{1}{2} e^{x^2} \sin(x^2) + C$

11. $x \sin(\ln x) + C$

12. $e^x \ln x + c$

13. $e^x \left(\frac{x-2}{x+2} \right) + c$

14. (3)

15. (3)

16. (2)

17. (3)

18. (4)

19. (2)

20. (2)

